

LABORATÓRIO 10

```
df = spark.read.parquet("hdfs://localhost:9000/user/hive/warehouse/silver_transactions")
df.show(5)
df.createOrReplaceTempView("silver_transactions")
```

Amostra carregada com as colunas:

transaction_id, customer_id, amount, transaction_type, status, risk_score,
is_fraud, ts, segment, credit_score, year, month, day, day_of_week, amount_band

Leitura da Silver e visualização.

```
spark.sql("""
SELECT segment, ROUND(AVG(credit_score), 1) AS score_medio, COUNT(*) AS total
FROM silver_transactions
GROUP BY segment
ORDER BY score_medio
""").show()
```

```
from pyspark.sql import functions as F
(df.groupBy("segment")
 .agg(F.round(F.avg("credit_score"), 1).alias("score_medio"),
      F.count("*").alias("total"))
 .orderBy("score_medio")
 .show())
```

Standard	643.9	29689
Premium	652.7	61156
High-Risk	659.8	9155

Os dois métodos retornaram o mesmo resultado.

Duas formas de consultar os mesmos dados pelo Spark.

```
spark.sql("""
SELECT dayofweek(ts) AS dia_semana,
       COUNT(*) AS total,
       SUM(CASE WHEN is_fraud THEN 1 ELSE 0 END) AS fraudes
FROM silver_transactions
GROUP BY dayofweek(ts)
ORDER BY dia_semana
""").show()
```

dia_semana	total	fraudes
1	14034	257
2	13940	283
3	14222	262
4	14338	234
5	14452	263
6	14484	261
7	14530	273